

Yiqi Zhao

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EDUCATION

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|---|---------------------------------------|
| University of Southern California
<i>PhD Candidate in Computer Science</i> <ul style="list-style-type: none">• GPA: 4.0/4.0• Supervisors: Prof. Jyotirmoy V. Deshmukh; Prof. Yue Wang• Viterbi School of Engineering Fellowship | Los Angeles, USA
2023/08 - Present |
| University of Southern California
<i>Master of Science in Computer Science</i> <ul style="list-style-type: none">• GPA: 4.0/4.0• Petitioned during PhD study. | Los Angeles, USA
2023/08 - 2025/08 |
| Vanderbilt University
<i>Bachelor of Science, Magna Cum Laude</i> <ul style="list-style-type: none">• GPA: 3.935/4.0• Majors: Honors in Computer Science; Mathematics• Minors: Electrical Engineering; Data Science• Dean's List; Good Standing | Nashville, USA
2020/08 - 2023/05 |

WORK EXPERIENCE

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| Research and Development Intern, Toyota Research Institute of North America
<i>Supervisors: Georgios Fainekos, Bardh Hoxha</i> <ul style="list-style-type: none">• Worked on Cyber-Physical Systems research at the Future Research Department.• Focused on Model Predictive Path Integral for Path Planning. | Ann Arbor, USA
2025/05 - 2025/08 |
| Application Development Intern, ADP
<i>Supervisor: Jim Farber</i> <ul style="list-style-type: none">• Full stack API development for the Autopay Group within the Global Product & Tech group. | Roseland, USA
2022/06 - 2022/08 |

RESEARCH INTERESTS

Robotics, Cyber-Physical Systems, Formal Methods, Systems and Control Theory

PREPRINTS (* INDICATES EQUAL CONTRIBUTION; † DENOTES EQUAL ADVISING)

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- [p5] **Yiqi Zhao**, Taekyung Kim, Hideki Okamoto, Bardh Hoxha, Jyotirmoy V. Deshmukh, Lars Lindemann, Georgios Fainekos. "Safety-aware Model Predictive Path Integral Control with Signal Temporal Logic". *Arxiv*, 2026. [[paper](#)]
- [p4] Celina Shiyu Wang, **Yiqi Zhao**[†], Junjie Ye, Yue Wang, Jyotirmoy V. Deshmukh[†]. "Logic-VLA: A Temporal Logic Conditioned Vision-Language-Action Model". *Arxiv*, 2026. [[paper](#)]
- [p3] **Yiqi Zhao**^{*}, Ruohai Ge^{*}, Celina Shiyu Wang, Junjie Ye, Muchen Xu, Minhao Li, Sergey Zakharov, Basile Van Hoorick, Vitor Campagnolo Guizilini, Leonidas Guibas, Gaurav S. Sukhatme, Jyotirmoy V. Deshmukh, Yue Wang. "DUET: Dual-Robot Understanding via Efficient Teaching". *Under review*, 2026. [[paper](#)] [[project](#)]
- [p2] **Yiqi Zhao**^{*}, Xinyi Yu^{*}, Matteo Sesia, Jyotirmoy V. Deshmukh, and Lars Lindemann. "Conformal Predictive Programming for Chance Constrained Optimization". *Arxiv*, 2025. *Currently under review in Automatica*. [[paper](#)] [[codes](#)]
- [p1] **Yiqi Zhao**, Ziyan An, Meiyi Ma, and Taylor Johnson. "EduSAT: A Pedagogical Tool for Theory and Applications of Boolean Satisfiability". *Arxiv*, 2023. [[paper](#)] [[tool](#)]

CONFERENCE PAPERS

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- [c3] Anand Balakrishnan, Rohit Bernard, Shreeram Narayanan, Vidisha Kudalkar, **Yiqi Zhao**, Parinitha Nagaraja, Georgi Markov, Christof Budnik, Helmut Degen, Lars Lindemann, Jyotirmoy V. Deshmukh. "Safety Assurance for Autonomous Systems with Multiple Sensor Modalities". *The 22nd ACM-IEEE International Symposium on Formal Methods and Models for System Design (MEMOCODE)*, Raleigh, USA. 2024. [[paper](#)]

[c2] **Yiqi Zhao**, Bardh Hoxha, Georgios Fainekos, Jyotirmoy V. Deshmukh, and Lars Lindemann. "Robust Conformal Prediction for STL Runtime Verification under Distribution Shift". *The 15th ACM/IEEE International Conference on Cyber-Physical Systems*, Hong Kong, China. 2024. **Best Paper Award Finalist**. [[paper](#)] [[codes](#)]

[c1] **Yiqi Zhao**, Ziyang An, Xuqing Gao, Ayan Mukhopadhyay, Meiyi Ma. "Fairguard: Harness Logic-based Fairness Rules in Smart Cities". *The 8th ACM/IEEE Conference on Internet of Things Design and Implementation*, San Antonio, USA. 2023. [[paper](#)]

JOURNAL PAPERS (* INDICATES EQUAL CONTRIBUTION.)

[j5] Lars Lindemann, **Yiqi Zhao**, Xinyi Yu, George J. Pappas, Jyotirmoy V. Deshmukh. "Formal Verification and Control with Conformal Prediction: Practical Safety Guarantees for Autonomous Systems". *IEEE Control Systems*. 2025. [[paper](#)] [[tutorial](#)]

[j4] **Yiqi Zhao***, Xinyi Yu*, Bardh Hoxha, Georgios Fainekos, Jyotirmoy V. Deshmukh, and Lars Lindemann. "STL-GO: Spatio-Temporal Logic with Graph Operators for Distributed Systems with Multiple Network Topologies". *ACM Transactions on Embedded Computing Systems; the ACM SIGBED International Conference on Embedded Software (EMSOFT 2025, ESWEEK)*, Taipei, 2025. [[paper](#)]

[j3] Xinyi Yu, **Yiqi Zhao**, Xiang Yin, and Lars Lindemann. "Signal Temporal Logic Control Synthesis among Uncontrollable Dynamic Agents with Conformal Prediction". *Automatica*. 2025. [[paper](#)] [[codes](#)] (**Regular Paper.**)

[j2] **Yiqi Zhao**, Emily Zhu, Bardh Hoxha, Georgios Fainekos, Jyotirmoy V. Deshmukh, Lars Lindemann. "Distributionally Robust Predictive Runtime Verification under Spatio-Temporal Logic Specifications". *ACM Transactions on Cyber-physical Systems*. 2025. [[paper](#)] [[codes](#)]

[j1] Ziyang An, **Yiqi Zhao**, Xuqing Gao, Ayan Mukhopadhyay, Meiyi Ma. "Formal Logic-Guided Harnessing Heterogeneous Fairness Rules in Smart Cities". *ACM Transactions on Cyber-physical Systems*. 2025. [[paper](#)]

NON-AUTHORSHIP ACKNOWLEDGED CONTRIBUTIONS

[o2] Junyang Cai, Weimin Huang, Jyotirmoy V. Deshmukh, Lars Lindemann, Bistra Dilkina. "Neuro-Symbolic Acceleration of MILP Motion Planning with Temporal Logic and Chance Constraints". *Arxiv*, 2025. [[paper](#)] (Acknowledged for the help.)

[o1] Hanwen Liu, Shashank Giridhara, Ibrahim Sabek. "Conformal Prediction for Verifiable Learned Query Optimization". *Proceedings of the VLDB Endowment*. 2025. [[paper](#)] (Acknowledged for the support.)

SELECTED HONORS

ICCPs Best Paper Award Finalist May. 2024
Awarded at the 15th ACM/IEEE International Conference on Cyber-Physical Systems for the paper "Robust Conformal Prediction for STL Runtime Verification under Distribution Shift".

Viterbi School of Engineering Fellowship Aug. 2023
Awarded for selected incoming PhD students at Viterbi School of Engineering, USC.

WISE Award (Vanderbilt Institute for Surgery and Engineering Summer Fellowship) 2021
Awarded for the paid fellowship at the Vanderbilt Institute for Surgery and Engineering Summer Fellows Program.

TEACHING EXPERIENCE

Teaching Assistant of CSCI 577A (Software Engineering), University of Southern California Los Angeles, USA

Supervisors: [Jae Young Bang](#), [Mahdi Eslamimehr](#)

2026/01 - 2026/05

- Graded students' homework and exams.

Grader of CS 4260 (Artificial Intelligence), Vanderbilt University

Nashville, USA

Supervisors: [Meiyi Ma](#), [Daniel Moyer](#)

2022/08 - 2023/05

- Graded students' homework and exams.

- Held TA office hours regularly.

ACADEMIC SERVICES

Journal Reviews for NAHS

Conference Reviews for ACC 2024, VMCAI 2024, CDC 2024, ICLR 2025, AISTATS 2025, HSCC 2025, L4DC 2025, ICCPS 2025, NEUS 2025, MECC 2025, ICRA 2026, AISTATS 2026, L4DC 2026, NeurIPS 2026.

Conference PC Member and Reviews for HSCC 2024 (Repeatability Evaluation), HSCC 2025 (Repeatability Evaluation), ICCPS 2025 (Posters and Demos).

Tutorial on Formal Verification and Control with Conformal Prediction at the CPS-IoT Week 2025 [[home-page](#)].
05/06/2025

SELECTED TALKS

The 15th ACM/IEEE International Conference on Cyber-Physical Systems	05/16/2024
The 43rd Southern California Control Workshop (with Xinyi Yu)	04/19/2024
The 8th ACM/IEEE Conference on Internet of Things Design and Implementation	May 2023

SKILLS

Programming Languages	Python, Javascript, C++, C, MATLAB, Java, SQL. Mandarin Chinese (native), English (TOEFL: 116).
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